MASTER'S THESIS
(PHYSICS)

Evolution of Open Quantum Systems under Continuous Measurement of their Environment

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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CONTEXT AND MOTIVATIONS

This document is the report of an internship carried out as part of a Master's degree in fundamental and numerical physics (CompuPhys).
My internship has been conducted within the team of Theorical Physics — *équipe de Physique Théorique* ΦTh — of the *UTINAM Institute* in the building of the *Observatory of Besançon*.
It aims to study from an academical perspective at first, the subject of continuous measurement considering open quantum systems. The main objective was to explore a subject almost-unknown in details to either my supervisor or myself. We wanted to understand the possibilities and how to handle this domain properly. This work could then lead to other searches linked to continuous measurements.

By familiarizing ourselves with the topic, we decided to adopt this perticular frame of report. The structure of this document is meant to help any unfamiliar reader (about continuous measurement), to discover and explore its fundamentals. We wanted to make it as illustrative and pedagogical as possible, thus, simple examples will be used to graphically depict mathematical and physical concepts.

Introduction

"Quantum mechanics is the description of the behavior of matter and light in all details and, in particular, of the happenings on an atomic scale" said *R.Feynman* in one of his lectures [1]. In this framework, position is no longer a single well-defined value prior to measurement, instead, position is described by a probability density all across space. Consequently, the notion of a trajectory as a continuous succession of points in spacetime is lost in quantum dynamics. The definition of trajectory will therefore be addressed, but one question remains paramount when mentioning systems: how can we even define a quantum system properly?

In the ideal case, systems are treated as closed, *i.e.*, isolated from any external environment and interacting with nothing at all. This modelling allows quantum mechanics to describe dynamics of systems with unitary and deterministic terms (the state at time t is determined solely by knowing the state at time t_0) using the Schrödinger equation. [2] However in the physical world, systems do "feel" their surroundings, *i.e.*, they interact with their environment. This interaction should be taken into account to obtain a more complete and physically relevant representation. In doing so, we describe the so-called *Open Quantum System* (OQS) theory [3].

Other useful theories have been developed through the last decades, allowing quantum mechanics' description to be more and more complete. One of them being the Quantum Measurement Theory (QMT) [4], where the concept of measurement is fully described. Measuring a system has a tremendous impact on the its quantum properties. Furthermore, since any experimental probe inevitably interacts with the system, it is crucial to figure out the nature of this interaction. Measuring a system without impacting its quantum properties leads to the concept of weak measurements, that we will develop further below.

By studying QMT, one can thus notice that measuring a quantum system alters it. The measurement usually interacts so strongly with the system that it collapses into a state — an eigenstate of the observable — that lost interesting quantum properties. For example it can lose its entanglement[1]. Along the same lines, quantum systems can be in superposition of states, taking *Schrödinger's cat* thought experiment, this corresponds formally to a superposition of 'alive' and 'dead' states prior to measurement. Measuring the state of the cat will determine its fate, alive or dead, it is then "stuck" in the state measured and cannot be written as a superposition again. The idea of **weak measurements** is to **measure the environment** coupled with the system. Instead of altering the state, we obtain pieces of information on the system. The measurements can even be conducted continuously, maximizing the amount of data the observer can get. These are the global ideas behind **Quantum Continuous Measurements** (QCM) [5].

Practically, this work will be organized as follows: the first section covers an academical progression from basics in quantum mechanics up to the aforementioned theories. A second section will cover the photodetection and homodyne detection of open quantum systems. The qubit and harmonic oscillator will be used to illustrate our point. The final part is the study of a system of interest on which we apply quantum continuous measurement. (Si on a le temps.)

[1] $|\psi\rangle_{AB} = \frac{|00\rangle+|11\rangle}{\sqrt{2}} \xrightarrow[\text{in space A}]{\text{measuring 0}} |00\rangle_{AB} = |0\rangle_A \otimes |0\rangle_B$ a product state (thus non-entangled).

I From closed systems to continuous measurements — in a nutshell

The examples and illustrations you can find in the following are made on `Julia` using `QuantumToolbox.jl` especially to simulate the dynamics.

I.i) Generalities on quantum mechanics

I.i.1⟩ States and time evolution

In quantum mechanics a physical system's state is described by a vector of a mathematical space, a Hilbert space:

$$|\psi\rangle \in \mathcal{H}$$

The evolution of a system depends on its energy, the Hamiltonian H is an hermitian[2] operator that accounts for the total energy of a system. The Schrödinger equation states it explicitly:

$$i\hbar \frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle$$

Instead of looking at the state at time t we can use an unitary[3] operator $\hat{U}$ such that, $|\psi(t)\rangle = \hat{U}(t,t_0)\,|\psi(t_0)\rangle$. From now on, we will omit the $\hat{}$ symbol over operators when it is obvious. We get,

$$\dot{U}(t,t_0) = -\frac{i}{\hbar} H(t) U(t,t_0)$$

One can solve this differential equation in general by using a chronological time-ordering operator [3]. Giving for the state evolution,

$$|\psi(t)\rangle = \mathcal{T}_{\leftarrow} \exp\left[-\frac{i}{\hbar}\int_{t_0}^{t} H(s)\mathrm{d}s\right] |\psi(t_0)\rangle$$

In simpler terms, when the Hamiltonian is independent of time,

$$|\psi(t)\rangle = \underbrace{e^{-\frac{iH(t-t_0)}{\hbar}}}_{U(t,t_0)} |\psi(t_0)\rangle$$

This evolution of the state through time is said **unitary and reversible** because the U is unitary, and, because this equation also describe the time reversed evolution $t \to t_0$: $|\psi(t_0)\rangle = U^\dagger(t,t_0)\,|\psi(t)\rangle$.

I.i.2⟩ Observables and purity of states

Any physical quantity is represented in quantum mechanics by an **observable** which is an operator whose eigenvectors form an orthonormal basis of the Hilbert space $\mathcal{H}$ [2]. For finite dimension Hilbert space, hermitian operators[4] are observables. In the following we will consider finite dimension, non-degenerate cases to illustrate but one can generalize [2].

[2] $\hat{H}^\dagger = \hat{H}$, where † is the conjugate transpose operation, defining the hermitian adjoint.

[3] $UU^\dagger = U^\dagger U = \mathbb{I}$, where $\mathbb{I}$ is the identity.

[4] Hermitian operators always form a basis in finite dimension; we can use Gram-Schmidt process to find its orthonormal counterpart.

A natural operator whose eigenbasis is particularly interesting is the Hamiltonian, its eigenvalues correspond to the possible energy levels of a system.
Let us call $\{\phi_n\}_n$ the family of orthonormal eigenvectors of H which forms a basis of $\mathcal{H}$. Since it forms a basis of the Hilbert space, one can describe any state in this very basis:

$$\forall\, |\psi\rangle \in \mathcal{H}; \qquad |\psi\rangle = \sum_{n=1}^{\dim(\mathcal{H})} \underbrace{c_n}_{\langle\phi_n|\psi\rangle} |\phi_n\rangle$$

This kind of description in a base is obviously valid for any observable, not only for the Hamiltonian.

I.i.3⟩ Postulates of quantum mechanics

Physicists developed a corpus of the main properties of quantum mechanics, here are summarized the so-called postulates of quantum mechanics [2]:

1. The state of a system is represented by a density matrix $\rho \in \mathcal{L}(\mathcal{H})$; for pure states we can describe them using $|\psi\rangle$ (seen in I.i.1⟩), pure density matrix becomes simply $\rho = |\psi\rangle\langle\psi|$.

2. Physical quantities $\mathcal{A}$ are associated with operators $\hat{A}$, called observables. In the context of finite dimension space, they are simply Hermitian operators.

3. The only possible outcomes of a measurement of $\mathcal{A}$ are the eigenvalues a of $\hat{A}$, defined by $\hat{A}\,|a\rangle = a\,|a\rangle$.

4. When measuring $\mathcal{A}$ on a normalized system $|\psi\rangle$, we know the probability to get one of the possible eigenvalues: $\mathbb{P}(a_n)$ (which depends on the spectrum of the observable).

5. If the measurement of the physical quantity $\mathcal{A}$ on the system in the state ψ gives the result a_n, the state of the system immediately after the measurement becomes the eigenvector ϕ_n.

6. The time evolution of the wavefunction $|\psi(t)\rangle$ is governed by the Schrödinger equation:
$$i\hbar\frac{\mathrm{d}\,|\psi(t)\rangle}{\mathrm{d}t} = \hat{H}(t)\,|\psi(t)\rangle$$
where $\hat{H}$ is the hamiltonian of the system.

The kind of dynamics a system follows is then unitary and deterministic. We can always solve, at least analytically, for the solution of the Schrödinger equation.

I.ii) Open Quantum Systems (OQS)

I.iii) Quantum Measurement Theory (QMT)

II Continuously measured quantum systems – obtaining a Stochastic Master Equation (SME)

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